UNIVERSIDAD DEL ISTMO DE GUATEMALA

FACULTAD DE INGENIERIA

ELECTRONICA 2 Y DISENO ELECTRICO 1

# PCIe PI7C9X3G808GP




KEVIN SEBASTIAN FLORES LOPEZ

OCTUBRE DE 2026

| | | |
|---|---|---|
| UNIS - Facultad de Ingeniería - Electronica 2 | | |
| Title: PORTADA | | |
| Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0 |
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# INDICE DE HOJAS


| Hoja | Pag. |
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| 02_INDICE DE HOJAS | 2 |
| 03_DIAGRAMA DE BLOQUES Y POTENCIA | 3 |
| 04_ENTRADA DE POTENCIA 12V/5V | 4 |
| 05_BUCK 1: 12V -> 3.3V | 5 |
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| 07_BUCK 3: 5V -> 0.95V VP | 7 |
| 08_LDO 0.95V C_VDDC / 1.8V | 8 |
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| 18_LEDS PORTGOOD | - |
| 19_SMBus / I2C | - |
| 20_NOTAS DE DISENO | - |


**HISTORIAL DE REVISIONES**

| Rev | Fecha | Descripcion |
|---|---|---|
| 0.1 | 24/10/2026 | Diseno base + 3 convertidores Buck propios (TI WEBENCH) |
| 0.2 | 28/10/2026 | Straps obligatorios y stack-up de 6 capas personalizado |
| 1.0 | 31/10/2026 | Esquematico reorganizado en 15 hojas (potencia y alta velocidad) |

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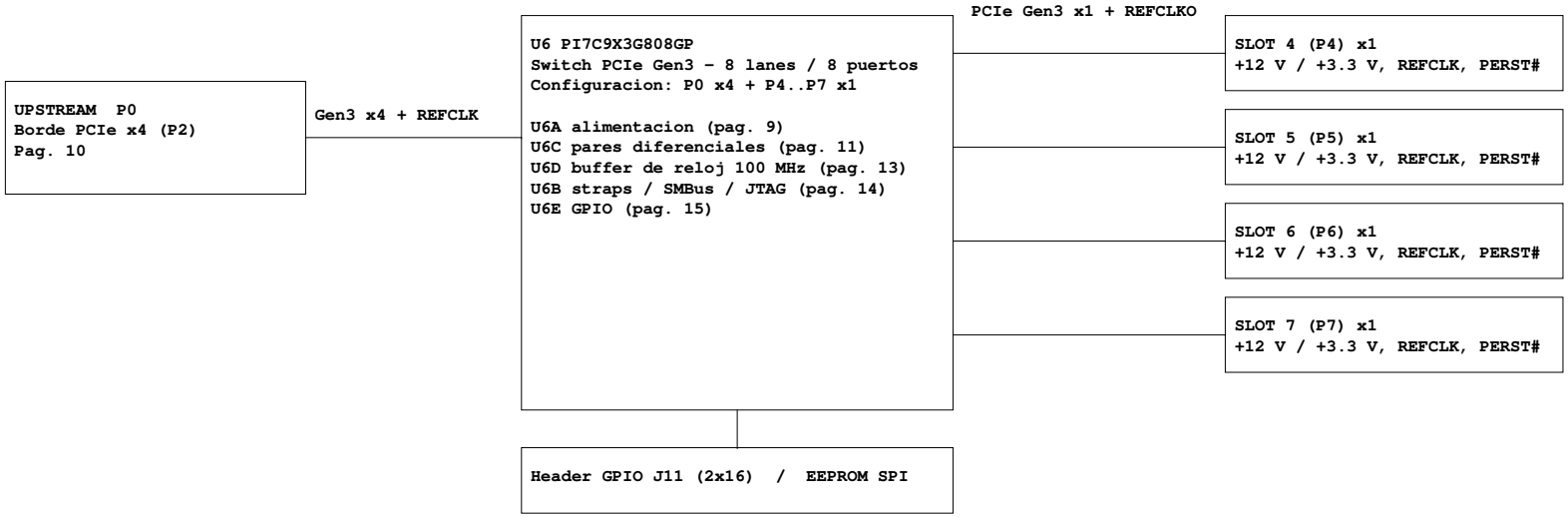
Title: INDICE DE HOJAS

Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0

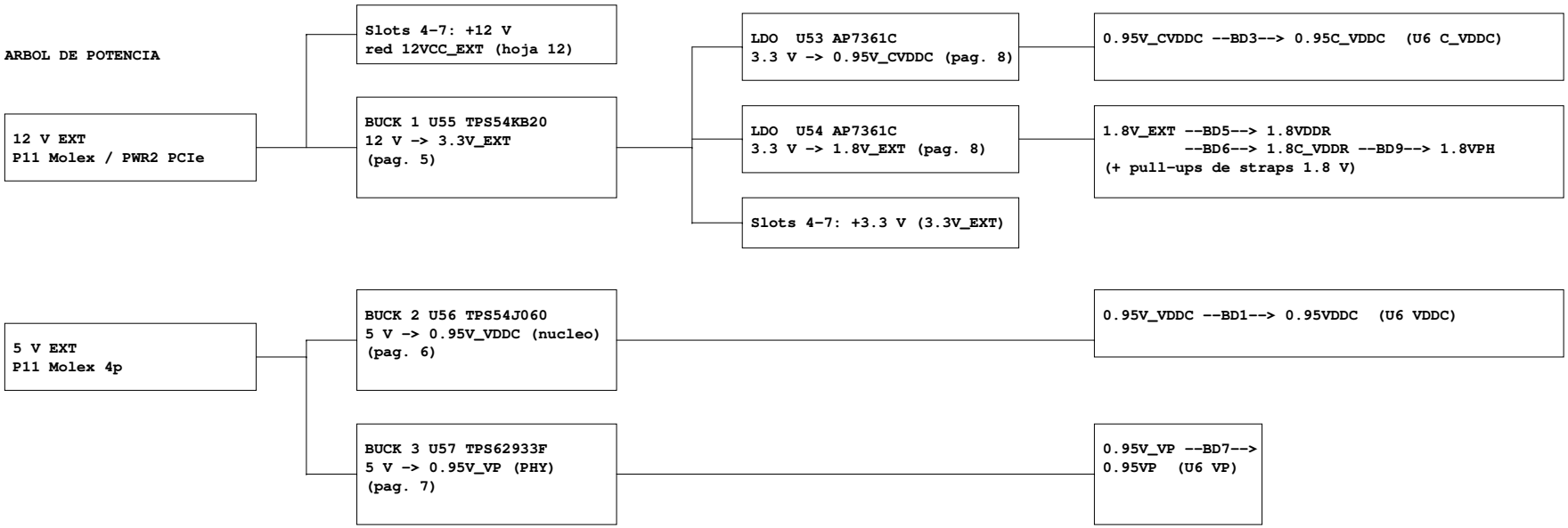
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# DIAGRAMA DE BLOQUES Y ARBOL DE POTENCIA

**DIAGRAMA DE BLOQUES (SENALES)**




**ARBOL DE POTENCIA**

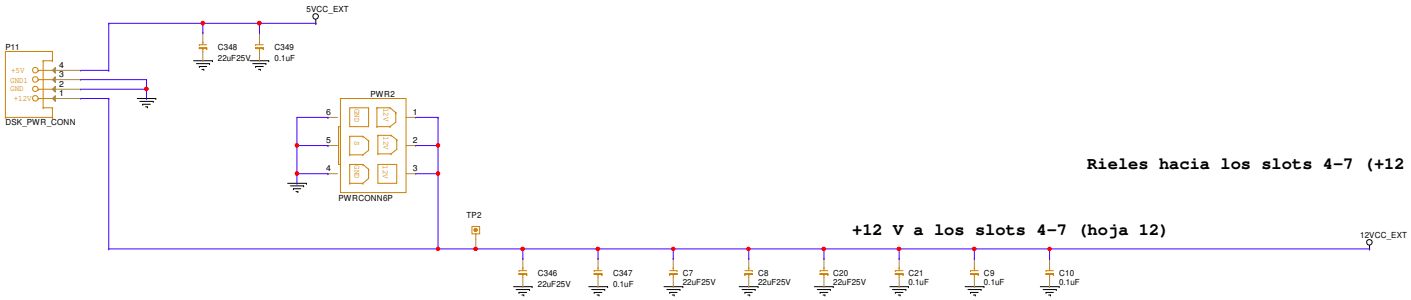


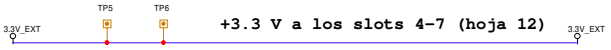

**Secuencia de arranque: 3.3V_EXT (PG_3V3) -> EN Buck 2 -> PGOOD (U2_PGOOD) -> EN Buck 3. LEDs indicadores en pag. 4.**

| | | |
|---|---|---|
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| Title: DIAGRAMA DE BLOQUES Y ARBOL DE POTENCIA | | |
| Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0 |
| Date: Saturday, October 31, 2026 | Sheet 3 of 15 | |

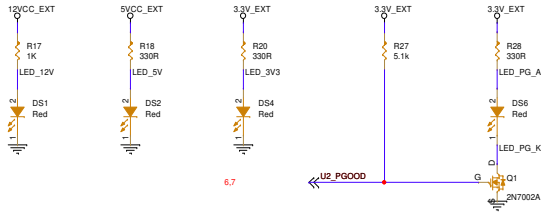
# ENTRADA DE POTENCIA 12 V / 5 V

**Entrada de 12 V y 5 V externos (P11 Molex 4p / PWR2 PCIe 6p), desacople bulk, distribucion a los slots 4-7 e indicadores LED.**



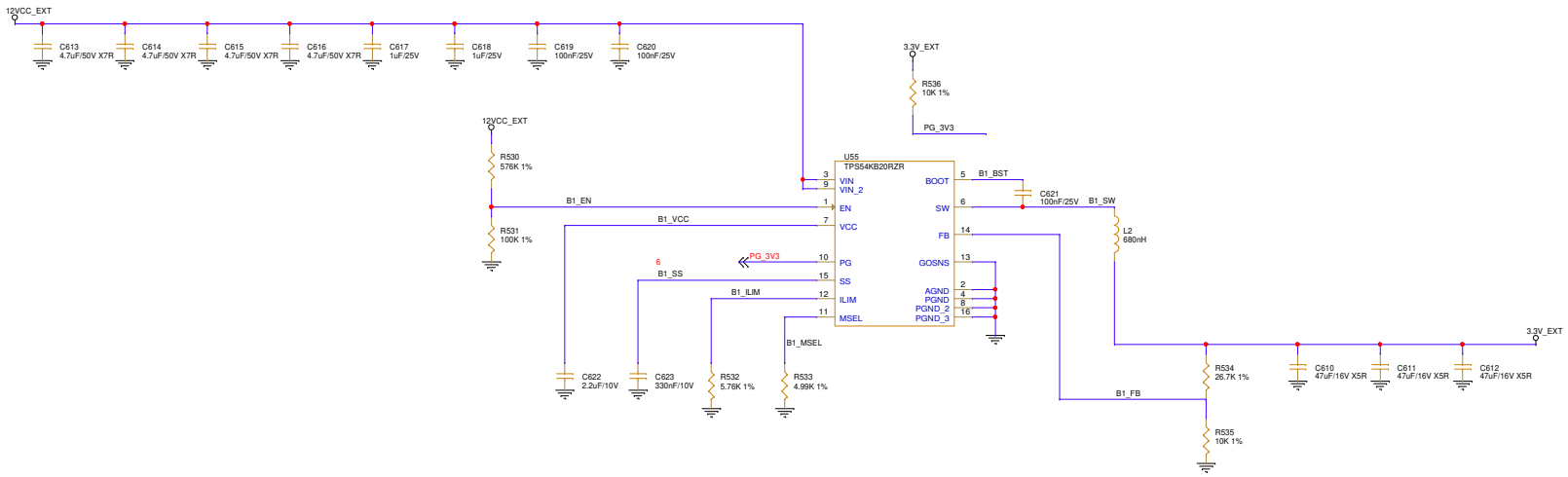

**Indicadores: LED por riel y LED de PGOOD de 0.95V_VDDC (Q1)**




| | | |
|---|---|---|
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| Title: ENTRADA DE POTENCIA 12 V / 5 V | | |
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# BUCK 1:  12 V -> 3.3 V

**U55 TPS54KB20: 12 V -> 3.3V_EXT. Alimenta los slots (+3.3 V) y los LDO U53/U54. Su PG (PG_3V3) habilita el Buck 2.**




| NOTAS DE DISENO – BUCK 1 (TPS54KB20, dimensionado en TI WEBENCH) |
|---|
| Vin = 12 V (10.8 a 13.2 V)  Vout = 3.3 V Iout = 15 A  fsw = 800 kHz (seleccion por MSEL, R533) |
| Vout = Vref x (1 + R534/R535) = 0.9 V x (1 + 26.7k/10k) = 3.303 V |
| UVLO de EN = 9 V (R530/R531)  Limite de corriente ~ 21 A (ILIM, R532)  Arranque suave ~ 8 ms (C623 = 330 nF) |
| L2 = 680 nH Coilcraft XAL8080 (37 A)  Cout = 3 x 47 uF X5R  Eficiencia 94.5 %  Rizado 6.8 mVpp |
| PG (drenador abierto, pull-up R536) -> PG_3V3 = EN del Buck 2 (hoja 6). GOSNS a GND: sensado local. |

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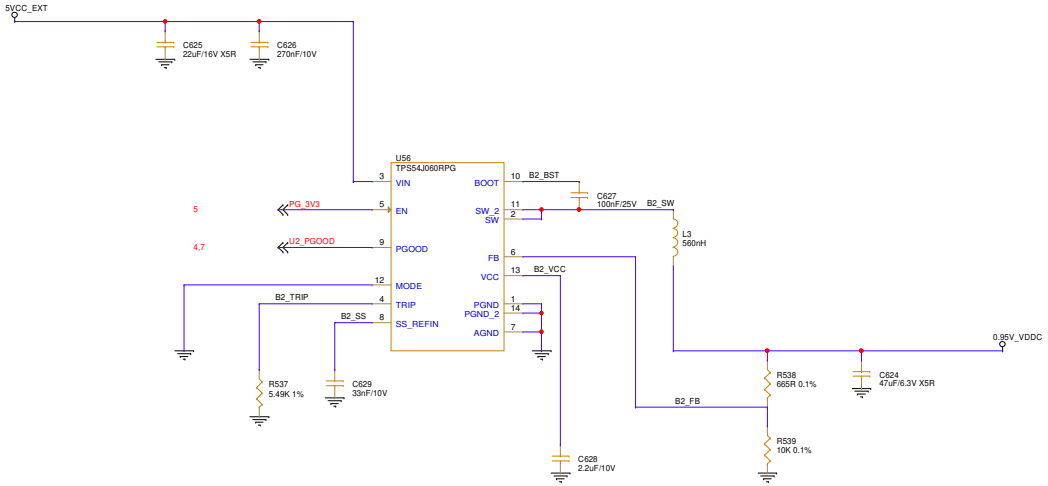
Title: BUCK 1: 12 V -> 3.3 V (TPS54KB20)

Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0

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# BUCK 2:  5 V -> 0.95 V VDDC

**U56 TPS54J060: 5 V -> 0.95V_VDDC (nucleo del switch). EN = PG_3V3. Su PGOOD (U2_PGOOD) habilita el Buck 3 y enciende DS6.**

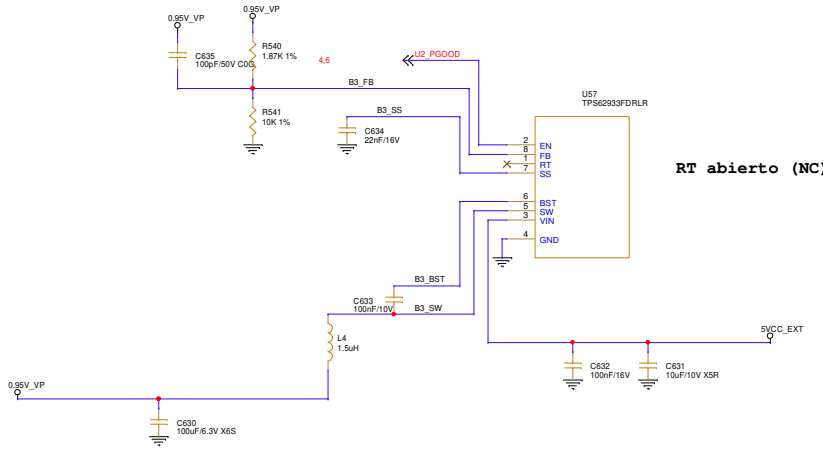



| NOTAS DE DISENO – BUCK 2 (TPS54J060, WEBENCH + ajuste propio del setpoint) |
|---|
| Vin = 5 V  Vout = 0.96 V  Iout = 4 A (VDDC del switch: 3.07 A max)  fsw = 1.1 MHz, FCCM (MODE a GND) |
| Vout = Vref x (1 + R538/R539) = 0.9 V x (1 + 665/10k) = 0.960 V  (resistencias de 0.1 %) |
| Setpoint de 0.96 V: compensa la caida en la ferrita BD1 (31 mV a 3.07 A) -> 0.90 a 0.99 V en las bolas del switch |
| Limite de corriente ~ 5.5 A (TRIP, R537)  Arranque suave ~ 3 ms (C629 = 33 nF)  L3 = 560 nH XAL5020 Ef. 84.9 % |
| EN = PG_3V3 (arranca despues del Buck 1). PGOOD -> U2_PGOOD = EN del Buck 3 (hoja 7) y LED DS6 (hoja 4). |

| | | |
|---|---|---|
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| Title: BUCK 2: 5 V -> 0.95 V VDDC (TPS54J060) | | |
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# BUCK 3:  5 V -> 0.95 V VP

**U57 TPS62933F: 5 V -> 0.95V_VP (PHY del switch). EN = U2_PGOOD. Vout = 0.8 V x (1 + R540/R541) = 0.95 V.**




**RT abierto (NC) = 500 kHz**

| **NOTAS DE DISENO – BUCK 3 (TPS62933F, dimensionado en TI WEBENCH)** |
|---|
| **Vin = 5 V  Vout = 0.95 V  Iout = 1 A (VP del switch: 0.57 A max)  fsw = 500 kHz, FCCM (PWM forzado)** |
| **Vout = Vref x (1 + R540/R541) = 0.8 V x (1 + 1.87k/10k) = 0.950 V  C635 = 100 pF de feed-forward** |
| **Pin RT abierto = 500 kHz (tabla de RT del datasheet; RT a GND = 1.2 MHz, resistor = 200 kHz a 2.2 MHz)** |
| **Arranque suave ~ 3 ms (C634 = 22 nF)  L4 = 1.5 uH XAL7030  Cout = 100 uF X6S  Ef. 89.4 % Rizado 3.0 mVpp** |
| **EN = U2_PGOOD: arranca despues del Buck 2 (secuencia VDDC -> VP).** |

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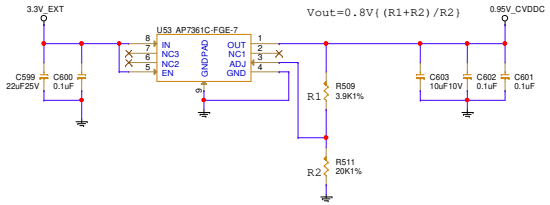
Title: BUCK 3: 5 V -> 0.95 V VP (TPS62933F)

Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0

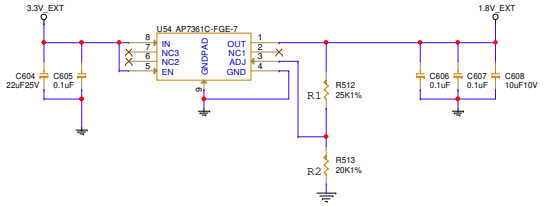
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# LDO 0.95 V C_VDDC / 1.8 V

0.95V / 1A

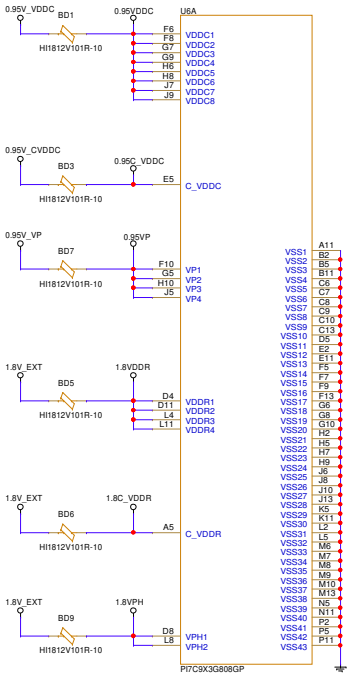


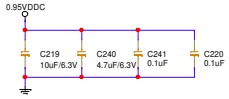

1.8V / 1A

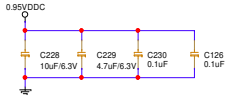


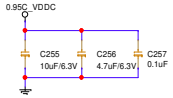

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| Title: LDO 0.95 V C_VDDC Y 1.8 V (AP7361C) | | |
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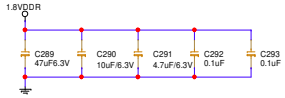
# U6A - ALIMENTACION Y DESACOPLE

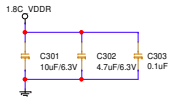


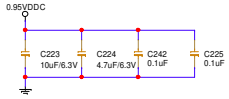



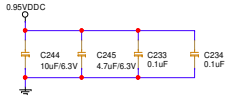



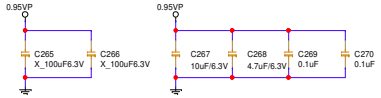



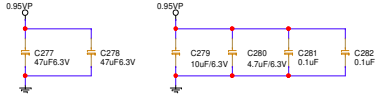



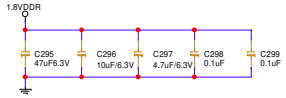



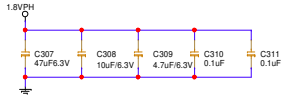



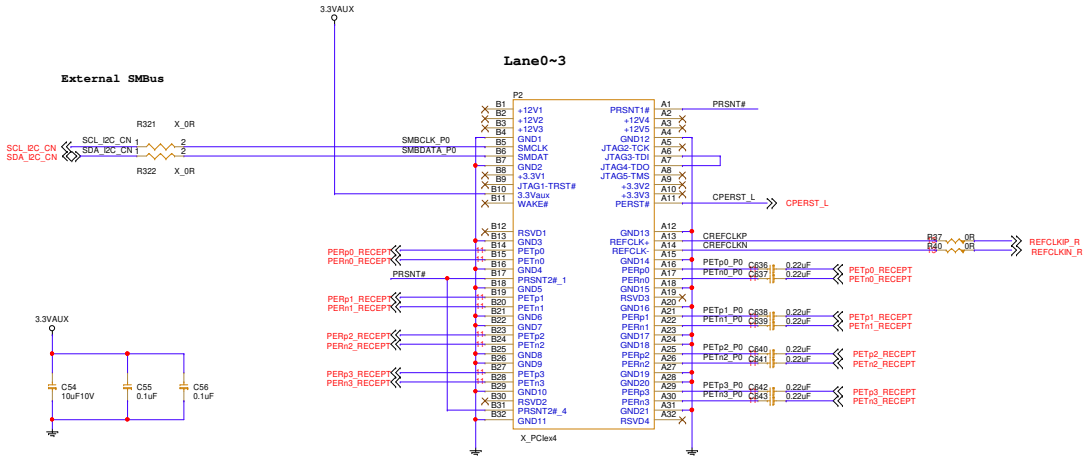



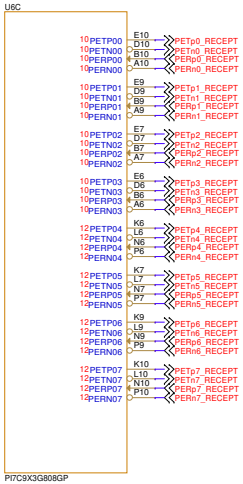

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| Title: U6A: ALIMENTACION Y DESACOPLE DEL SWITCH | | |
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# UPSTREAM PCIe x4 (PUERTO 0)



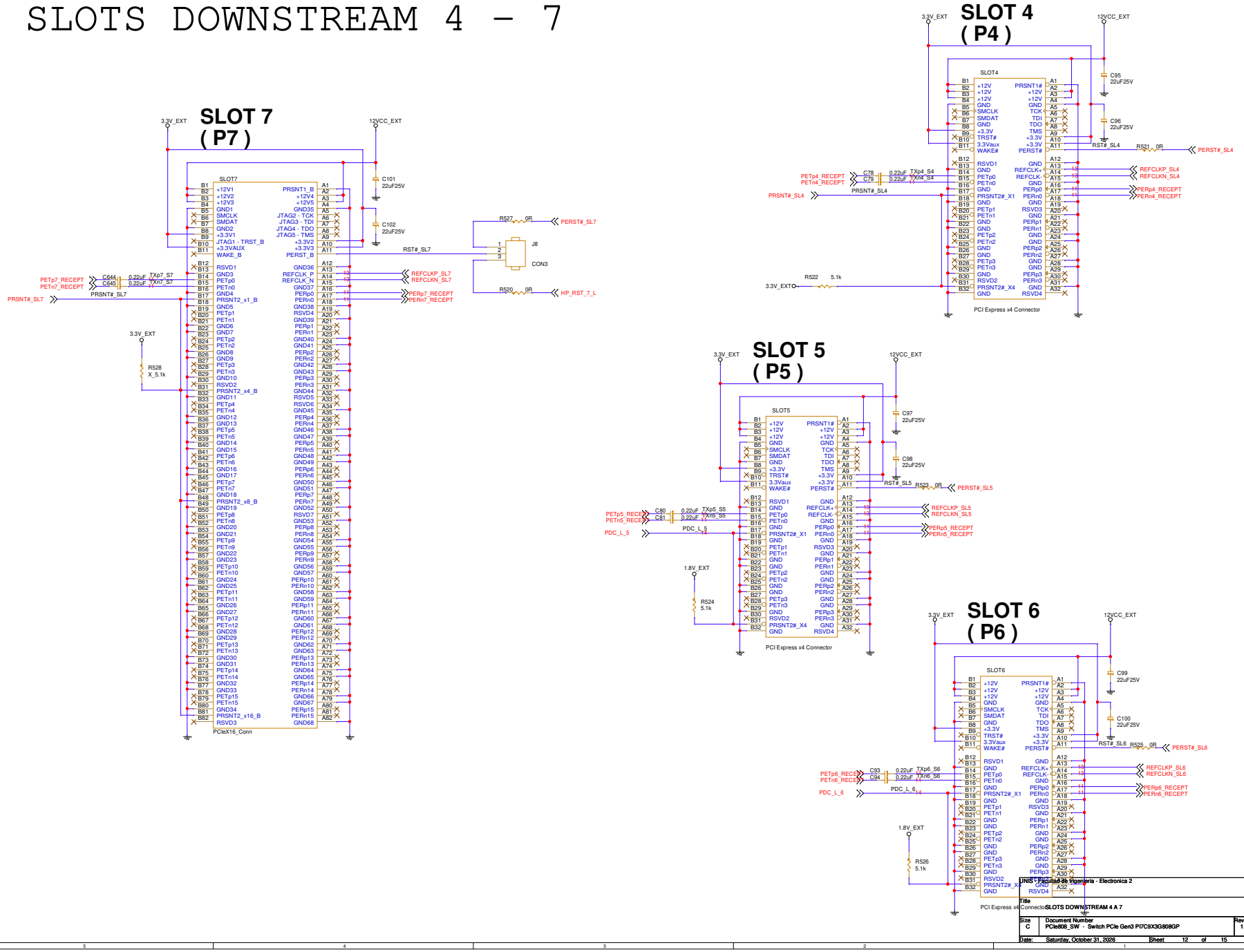

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|---|---|---|
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| Title: UPSTREAM PCIe x4 (PUERTO 0) | | |
| Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0 |
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# U6C - PARES DIFERENCIALES

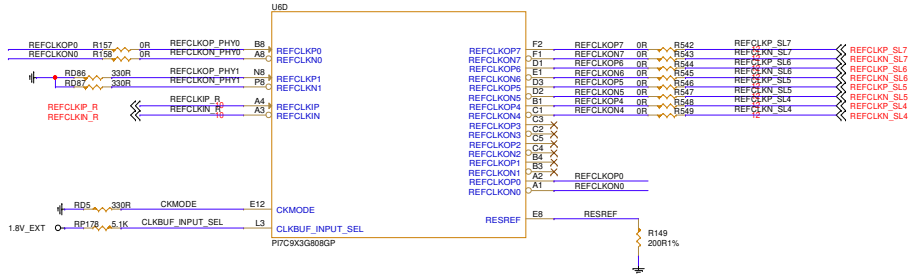



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|---|---|---|
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| Title: U6C: PARES DIFERENCIALES PCIe | | |
| Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0 |
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# SLOTS DOWNSTREAM 4 - 7



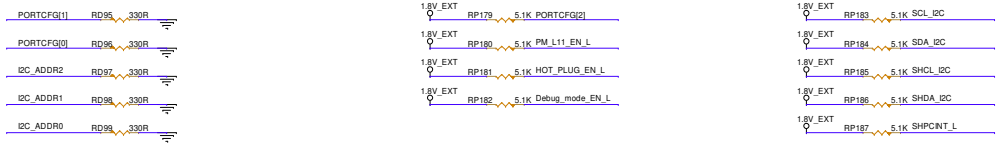
# U6D – RELOJ DE REFERENCIA



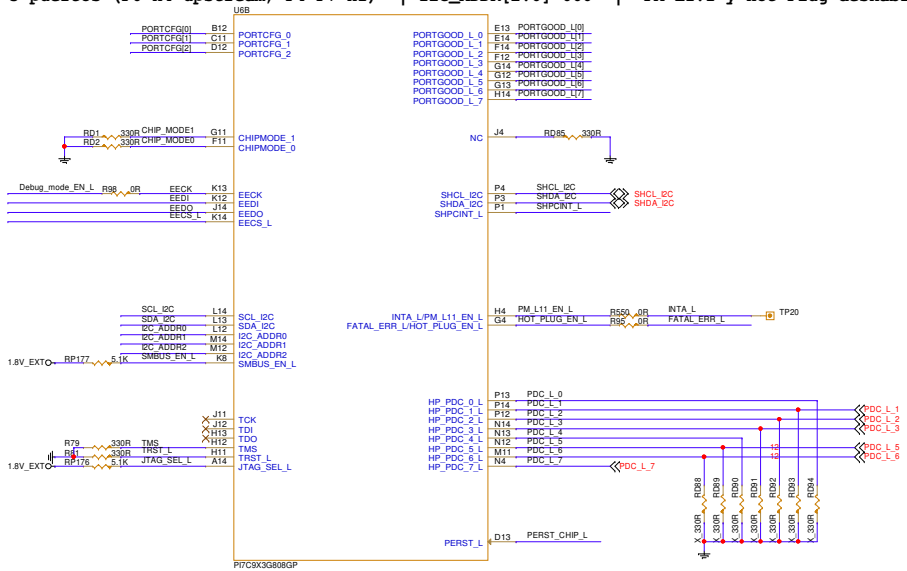

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|---|---|---|
| UNIS - Facultad de Ingeniería - Electronica 2 | | |
| Title: U6D: RELOJ DE REFERENCIA | | |
| Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0 |
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# U6B – CONFIGURACION Y CONTROL

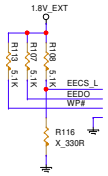
**STRAPS DE CONFIGURACION (obligatorios, ver DS43471 sec. 3.2/3.5)**




**PORTCFG[2:0]=100b: modo 5 puertos (P0 x4 upstream, P4–P7 x1) | I2C_ADDR[2:0]=000 | PM L1.1 y Hot-Plug deshabilitados | Debug mode deshabilitado**



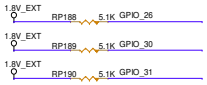

**PULL-UPS DE LA INTERFAZ EEPROM SPI. R116 = DNP**

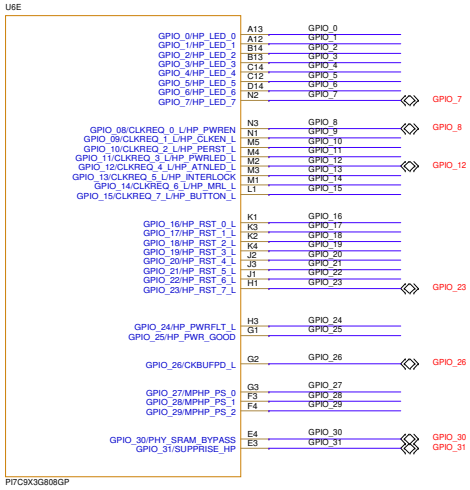


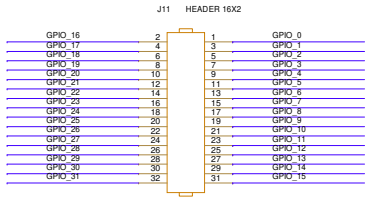

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|---|---|---|
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| Title: U6B: CONFIGURACION, STRAPS, SMBus Y JTAG | | |
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# U6E - GPIO

**PULL-UPS DE GPIO CON FUNCION ALTERNA: GPIO26=CKBUFPD_L (buffer de reloj activo), GPIO30=PHY_SRAM_BYPASS, GPIO31=SURPRISE_HP**




| | | |
|---|---|---|
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| Title: U6E: GPIO Y HEADER J11 | | |
| Size: C | Document Number: PCIe808_SW - Switch PCIe Gen3 PI7C9X3G808GP | Rev: 1.0 |
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